Jessica Jeanne

Software Engineer · Linguist · Audio Engineer

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Work Experience

Software Engineering

PathCare — *Cape Town*

Frontend Developer — *Apr 2023 - Feb 2025*

- Worked on rebuilding a mobile app using Angular and Ionic; worked with product owners to implement multiple pages as well as custom components for improved user experience.
- Improved accessibility of the mobile version of a website by making it still legible and functional for users with poor eyesight using a very large font on a mobile device.
- Improved processes by introducing a code review tool, making code reviews more efficient.

Languages and Skills

Angular, TypeScript/JavaScript, CSS, CouchDB, Jenkins, JetBrains Space, MS SQL, Java

Sozo Labs — *Cape Town*

Full Stack Developer — *Apr-Sep 2021*

- Improved existing system (web frontend in Angular and app backend in C# Xamarin), streamlining code, bug fixes, and adding features.
- Moved the system onto AWS, wrote documentation on best practices for setting up in AWS, and made use of Cloud Formation.
- Set up a user sign-up system with Amazon Cognito which made use of Amazon SES and AWS Lambda.

Languages and Skills

Angular, React, TypeScript, C#, Xamarin, .NET, AWS Cloud Formation, AWS Lambda, Amazon Cognito, MySQL, Docker, Bitbucket

Amazon Web Services — *Cape Town*

Software Development Engineer — *Aug 2016 - Sep 2019*

- Under the guidance of a senior engineer I worked on moving APIs from a legacy networking framework to a newer networking framework, which gave multiple services an increase in efficiency in their network modules.
- Took the lead for my team's contribution to add a new API to the EC2 services (send-diagnostic-interrupt). This involved high level project design around a micro-service architecture, and coordinating with other teams in Cape Town and Seattle.
- Participated in an operational support rotation which involved: responding to critical issues; server maintenance; co-ordinating daily integration, testing, and deployment of the service; and using a monitoring system built on Grafana to analyse metrics.
- Demonstrated attention to detail: the products worked on have a lot of working parts, and any new piece of code could have unintended consequences. A keen eye for detail helps to avoid this, or find existing problems in old pieces of code. This was one of my strongest areas.
- An internal wiki is maintained to provide visibility on teams and the products they own. I made significant contributions to this, updating and re-formatting the wiki that was out of date for one team, and correcting errors and maintaining the status of the wiki for the second team I was part of.

Languages and skills

Java, Ruby, AWS, system administration and DNS management, Java design patterns, integration testing, CI/CD, Agile workflows and project management, Git, teamwork across locations and time zones

Amazon Web Services — *Cape Town*

Software Development Engineer Intern — *Feb-Mar 2013 and Dec-Jan 2014-2015*

- Added a page to a server monitoring website built on Ruby on Rails, enabling users to schedule maintenance on servers.
- Contributed to an in-house reporting tool in Ruby, and got a service onboarded to the tool using YAML.

Languages and Skills

Ruby on Rails, YAML, Git, unit testing and test-driven development

Linguistics

National English Literary Museum — *Grahamstown*

Data Capture — *2012*

- Gathering data from online news articles to build a corpus of South African English from 2011, using a sampling frame from the BE06 corpus of British English.

Rhodes University Department of Linguistics — *Grahamstown*

Programmer — *Sep 2012*

- Used the E-Prime Psychology Software Tool to create a reaction-test (using a hardware device for accurate reaction timing) for a language researcher from the Linguistics department to use in her research.

Teaching & Tutoring

Rhodes University Computer Science Department
Grahamstown

Graduate Assistant (Tutor) *2012 – 2014*

- Tutoring groups of undergraduates (from first through third year) in Computer Science. It required being available for students in a support capacity, and included marking and, as part of that, managing mark-sheets.

Private
Grahamstown

Play therapy *Jul 2009 – Jul 2010*

- One-on-one play therapy with 2-3 year old autistic child.

Audio Engineering

Skills

Audio engineering for live bands, solo artists, speakers, and plays. Event and venue management. PA system setup.

National Arts Festival
Grahamstown

Sound Technician *Jul 2012*

- Venue management and sound engineering for the solo artist music venue, Graham Hotel, during the 11 days of the festival.

Rhodes Live Music Society
Grahamstown

Sound Engineer *2012, 2014*

- Setting up and engineering sound system equipment for live music events organised by the society.

Admin

Admin Assistant
Cape Town

Union Chapel Presbyterian Church *2022 - present*

- Database work using Airtable: Cleaned up and maintained the database of church members and volunteers. Researched and built a database of homeless shelters and services in Cape Town to be used to appropriately direct people in need.
- Organised and managed volunteers with weekly duties.

Projects

Media catalogue website

Design, development, and deployment

- Built a fronted and backend in Dart with Flutter and Dart Shelf, to catalogue media items like books and DvDs.
- Implemented fuzzy searching using PostgreSQL.
- Set up a server on my local network and deployed using Jenkins.

Languages and skills

Dart, Flutter, PostgreSQL, RESTful web development, Jenkins

Self-hosted mailserver

System administration

- Bought a domain and server space, set up the DNS in Digital Ocean, and built a custom mailserver initially from scratch to learn the details of how a mailserver works, and then using Mailu and Docker.

Languages and skills

Mail components (Dovecot, Postfix, certificate management), Docker, DNS management

Various mobile apps for personal use

Design and development

- I have designed and developed a few apps for personal use; namely a music player app that also has a desktop version, a simple anxiety symptom tracker, and a prayer list.

Languages and skills

Dart, Flutter, SQLite

Papers and Presentations

Social Media Mining for Health (SMM4H)

Paper for shared task <Relevance-weighted Sentiment Analysis of Reddit Posts> *2024*

- Participated in a shared task to analyse and classify the sentiment of reddit posts from a social anxiety subreddit

Southern Africa Telecommunication Networks and Applications Conference (SATNAC)

Presenter for <An Ethernet AVB Effects Unit Implementation Using XMOS Architecture> *2015*

- Updated and full presentation of the work I was doing for my masters research, using the XMOS AVB microcontroller as a hardware plug-in for audio processing

Southern Africa Telecommunication Networks and Applications Conference (SATNAC)

Poster presentation <Using External Ethernet AVB Devices for Audio Plug-in Processing> *2014*

- Work-In-Progress poster presentation of the work I was doing for my masters research, using the XMOS AVB microcontroller as a hardware plug-in for audio processing

Education

Bachelor of Arts, Rhodes University *Grahamstown*

BA in Computer Science and Combined English Literature and Linguistics *2009-2011*

CS Languages and skills: Python, Java, C++, OOP, algorithms, data structures, compilers

Bachelor of Science (Honours), Rhodes University *Grahamstown*

BSc (Honours) in Computer Science, coursework and dissertation *2012*

Honours thesis: Notation systems for machine translation of sign languages

Honours thesis skills: Natural language processing with the Python NLTK

Honours coursework modules: AI, Hardware Interfacing, Computational Linguistics, Networks, Image Processing, Information Security

Incomplete - Master of Science, Rhodes University *Grahamstown*

MSc in Computer Science by dissertation *2013-2016*

Masters thesis: Audio processing over an audio network using AVB and XMOS

Masters thesis skills: C++, C, and XC; programming microcontrollers (XMOS); audio networking with AVB; audio processing with VSTs; research and technical writing

Occasional Studies, University of Cape Town *Cape Town*

Intro to Psychology *2025*

Skills Summary

Programming languages	TypeScript/JavaScript, Dart, Java, MySQL, PostgreSQL, SQLite, C++, C and XC, C#, Python, Ruby, MatLab
Frameworks and tools	Angular, Flutter, React, CouchDB and PouchDB, Xamarin, Svelte, AVB, E-Prime
Programming related	Git, unit testing and integration testing, CI/CD, Agile workflows, code review
Cloud and related	AWS, Docker, DNS management, system administration, Google Cloud
Concepts	RESTful web dev, Java design patterns, NLP, machine learning, audio processing with VSTs, hardware interfacing
Audio engineering skills	Live sound mixing, recording and editing audio, venue management, PA system setup
Personal skills	Teamwork across locations and time zones, mentoring and tutoring, research and technical writing